

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

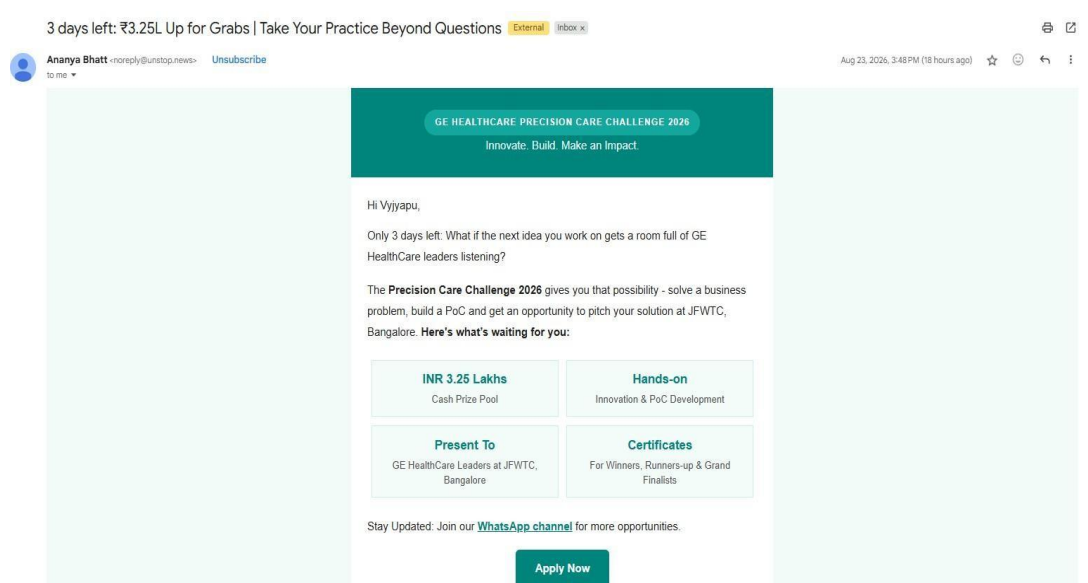
- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

1. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework)/ DKIM (DomainKeys Identified Mail):** Indicators of email authentication.



4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

Original Message

Message ID	<010201a02e20b56d-e6733248-3dbd-478d-866e-73b105a84d1d-000000@eu-west-1.amazonaws.com>
Created at:	Sun, Aug 23, 2026 at 3:48 PM (Delivered after 1 second)
From:	Ananya Bhatt <noreply@unstop.news>
To:	99240041232@klu.ac.in
Subject:	3 days left: ₹3.25L Up for Grabs Take Your Practice Beyond Questions
SPF:	PASS with IP 54.240.6.105 Learn more
DKIM:	'PASS' with domain unstop.news Learn more
DMARC:	'PASS' Learn more

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

5. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.

Fail: Indicates possible spoofing or tampering.

```

wAhVvYxLhAcQ9h3jv+OgSofB6_mW9hQKtAotg3g+1o59qQ0t1tBiE5UafwKnr2uThG7+HynAiEwd_vZvOn9bbAemM1t16hbE_TuWZouqT+1229w1Zy3_fg8HPV87tkq7Fjip5xx4NDc&body=Hi!
marketing_emails_from_your_company.
List-Unsubscribe-Post: List-Unsubscribe=One-Click
X-MC-Metadata: {"cid": "6a883eef8ee5159e9670946d", "user_id": "688f61322c5a20062eb5db18", "app_key": "BU64jRRiOTIBVKPC04ONDE61", "formatted_cid": "6a883eef8ee5159e9670946d", "to_email": "992400412320klu.ac.in"}
X-MSYS-API: {"campaign_id": "6a883eef8ee5159e9670946d", "metadata": {"cid": "6a883eef8ee5159e9670946d", "user_id": "688f61322c5a20062eb5db18", "app_key": "BU64jRRiOTIBVKPC04ONDE61", "formatted_cid": "6a883eef8ee5159e9670946d", "to_email": "992400412320klu.ac.in"}, "subject": "3 days left: U20b93.25L up for Grabs | Take Your Practice Beyond Questions", "formatted_cid": "6a883eef8ee5159e9670946d", "to_email": "992400412320klu.ac.in"}
Message-ID: <010201a02e0b5de-e6733248-3dbd-478d-866e-73b105a84dd1-000000@eu-west-1.amazonaws.com>
Date: Sun, 23 Aug 2026 10:18:10 +0000
Feedback-ID: By_FAK_N8cypLgsp.D1qWqYzXhA8usJsr_WgWjbuqjB9Z7zqrS3pCPCr8P5CnrrHrBnTQ5Q:6pF1DdD.XF.qg8_fzeFQ3o.Gh2B3acFxP8ix1.VcDfwKAK6hmnIAGLp8uCBoC0.mZjH_D.1.Ooq4W4G89E1q9h7WXkCQoAVD5t34C8ib0tw91pY8=:AmazonSES
X-SES-Outgoing: 2026.08.23-54.240.6.105

-----274782530174750310==
Content-Type: text/html; charset=utf-8"
MIME-Version: 1.0
Content-Transfer-Encoding: base64

```

7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.



[Pricing](#)
[Tools](#)
[Delivery Center](#)
[More](#)

[SuperTool](#)
[MX Lookup](#)
[Blacklists](#)
[DMARC](#)
[Diagnostics](#)
[Email Health](#)
[DNS Lookup](#)
[Analyze Headers](#)

SuperTool Beta9

[MX Lookup](#)

mx:unstop.news

Find Problems

Solve Email Delivery Problems

mx

Pref	Hostname	IP Address	TTL		
1	aspmx.l.google.com	64.233.180.27 Google LLC (AS15169)	5 min	Blacklist Check	SMTP Test
1	aspmx.l.google.com	2607:f8b0:4004:c1f::1b	5 min	Blacklist Check	
5	alt1.aspmx.l.google.com	192.178.234.26 Google LLC (AS15169)	5 min	Blacklist Check	SMTP Test
5	alt1.aspmx.l.google.com	2800:3f0:4003:c14::1b	5 min	Blacklist Check	
5	alt2.aspmx.l.google.com	172.253.116.26 Google LLC (AS15169)	5 min	Blacklist Check	SMTP Test
5	alt2.aspmx.l.google.com	2a00:1450:400b:c02::1a	5 min	Blacklist Check	
10	alt3.aspmx.l.google.com	192.178.223.26 Google LLC (AS15169)	5 min	Blacklist Check	SMTP Test
10	alt3.aspmx.l.google.com	2a00:1450:4009:c0f::1b	5 min	Blacklist Check	
10	alt4.aspmx.l.google.com	172.253.157.26 Google LLC (AS15169)	5 min	Blacklist Check	SMTP Test
10	alt4.aspmx.l.google.com	2a00:1450:400c:c23::1a	5 min	Blacklist Check	

Test

Result

Super tool Beta

amazonses.com MX Lookup

mx:amazonses.com Find Problems Solve Email Delivery Problems 

 Outlook.com has new sender requirements! [Learn More](#)
Get ready with MxToolbox Delivery Center!

Pref	10
Hostname	feedback-smtp.us-east-1.amazonses.com
IP Address	18.235.76.96 Amazon.com, Inc. (AS14618)
TTL	15 min
	Blacklist Check SMTP Test

Status	
Name	DNS Record Published
Response	DNS Record found

Status	
Name	BIMI Record Published
Response	Brand Logo not appearing in inboxes
	More Info

Status	
Name	DMARC Record Published
Response	DMARC Record found

Your IP is: 103.238.230.130 | [Contact](#) [Terms & Conditions](#) [Security](#) [API](#) [Privacy](#) Phone: (866)-698-6652 | © Copyright 2004-2026, MxToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

10. Document & Report Findings

Summary

- Sender: noreply@unstop.news
- Delivered via: Amazon SES (54.240.6.105)
- SPF: **Pass**
- DKIM: **Pass** (unstop.news, amazonses.com)
- DMARC: **Pass** (p=REJECT)
- **No anomalies detected**
- **Conclusion:** This email appears to be legitimate.

If suspicious activity is detected in any email,
report it to your IT department or email provider
for further investigation.



Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])
by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: Successfully analyzed email headers and identified phishing or email spoofing attacks using Mail Header Analyzer (MHA) or MX Toolbox Email Header Analyzer.